

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 17 September 2026, 8:12 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:13 AM**Time taken** 1 min 35 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main() {
4      int num;
5      int count = 0;
6
7      // Input: A positive integer num
8      if (scanf("%d", &num) != 1) {
9          return 0;
10     }
11
12     count++; // 1 for the loop initialization (i = 1)
13     for (int i = 1; i <= num; ++i) {
14         count++; // 1 for the condition check (i <= num)
15
16         if (num % i == 0) {
17             count++; // 1 for the printf execution inside the if statement
18         }
19
20         count++; // 1 for the loop increment (++i)
21     }
22
23     // Output: Print the value of the counter variable
24     printf("%d", count);
25
26     return 0;
27 }
28
```

	Input	Expected	Got	
✓	12	31	31	✓

	Input	Expected	Got	
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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